

Week 1 Project - E-commerce EDA Business Insights

This document summarizes the key observations from the E-commerce EDA challenge in a concise business-report format.

Insight 1: The dataset contains a large number of transactions, making it suitable for identifying meaningful business patterns.

Insight 2: Missing values are present in multiple columns and should be handled before model building.

Insight 3: Pricing-related variables are positively skewed, indicating that a small number of high-value purchases contribute significantly to revenue.

Insight 4: Product returns represent a meaningful business concern and should be monitored closely.

Insight 5: Quality-related issues and expectation mismatches appear to be major drivers of customer returns.

Insight 6: Some categorical features contain many unique values and require proper encoding before machine learning can be applied.

Insight 7: Outliers exist in price and order-value fields and should be considered during preprocessing.

Insight 8: Customer return behavior is likely influenced by multiple factors rather than a single feature.

Insight 9: A structured preprocessing pipeline can improve data quality and model performance.

Insight 10: Improving product descriptions and quality control may help reduce future return rates.

Conclusion: The exploratory analysis highlights several opportunities for business improvement and prepares the dataset for predictive modeling.